Tactix current N1–N4 audit
Public controller and proof/model checkpoint

Development report for Andrew

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Outcome

The current code-bearing checkpoint has a coherent public action/controller surface and passes its focused proof/model checks. A 46-test Debug matrix passes 46/46 in 220.04 seconds, the proof-oracle unit suite passes 69/69 in 11.035 seconds, and the existing complete Debug authority passes 285/285 in 957.60 seconds. The checkout is `9c695b479`; its solver, API, tests, and tools are byte-identical to code checkpoint `ec7e0ac86`. The audited Debug binary SHA-256 begins `51d82fe3d0395425`.

This is a current audit, not universal N1–N4 acceptance. It establishes strong scoped N1/N2 evidence. N1/N2 still require broad effective-policy coverage, N3 requires the current four regression gates, and N4 retains open cumulative and new-action scheduling evidence. No benchmark or performance campaign was launched, and the saved `d170b4a5` comparison was not rerun.

Requirement	Current evidence and remaining boundary	State
N1	39 public actions, exact branch phases, typed v5/v6/v7 theory payloads, controller offers, replay, and root/search boundaries pass focused tests; all reachable states and variable-length tuples remain non-exhaustive	pending
N2	Focused model reconstruction, certificate derivations, proof routing, real checker paths, corrupt-proof controls, and 69 oracle tests pass; effective both-policy and target/gate checking remain open	pending
N3	The saved baseline leads the four compared solvers, but the changed winner-integration tree has not passed B28+C5, Rank1–50, WV100, and WV2–89	pending
N4	Configuration ownership, execution signatures, rule/policy separation, and controller retention pass; ledger acceptance and new-action scheduling costs remain open	pending

N1 public applicability

The append-only registry contains 39 proof rules and policy operations. `api_n1`, `api_instances`, and `online_all_actions` pin the registry, action kind, declared instance surface, typed configuration, and action boundary. The Decide fixture requires both phases of every unassigned variable, including retired identities, and keeps off-fixpoint decisions legal. Vivify, Factor, and Subsume have separate noncanonical instance tests.

Variable-length theory requests use their actual typed interfaces. V5 applies GF(3) affine and DirectCSP requests; V6 deep-copies and applies one exact authenticated Gaussian batch; V7 does the same for an exact Tseitin connected component. Direct PHP uses the ordinary typed step and is terminal.

Controller tests cover exact collections, native offerers, restart ownership, retained trail semantics, random choice, deterministic replay, and failure-closed configuration.

The generic action fixture uses representative legal states. It does not prove every reachable state or tuple, and its scalar placeholder for V6/V7 is only a registry check; the dedicated v6/v7 tests own the real effects. Effective fuzz is therefore still required. Tseitin's physical-isolation restriction is declared in the X16 calculus precondition rather than hidden in policy.

N2 proof and model correctness

The focused run includes proof serialization, SR encoding, reconstruction, compacted-model validation, startup parity, clause-count preservation, Congruence SR, micro proofs, Sweep proofs, and certificate/public tests for the new theory actions. Tseitin independently checks ASCII and binary DSR and a repeated-root two-action proof through the proof oracle. Direct PHP validates its complete RAT/RUP event stream and exercises parser-finalized DSR publication. GF(3) Gaussian independently verifies each RUP event and the small-domain semantics before testing public DSR publication. Unsupported formats and corrupt or stale identities decline before mutation.

The 69 proof-oracle unit tests cover DSR/PBP/DRAT/LRAT routing, binary handling, every-lemma audits, duplicate-input bridges, deliberate lies, truncation, and known-bad proof rejection. These validate the checker machinery but do not generate every policy/rule path, so they cannot close universal N2 alone.

Performance and scheduling status

The saved five-arm Tier-D campaign remains strong historical evidence: Tactix has 986 independently verified solves and leads the four compared baselines in checked coverage and PAR-2. The current integration tree differs from that accepted no-MAB checkpoint. Its intended DefaultPolicy has not rerun the four regression gates, and the fixed 44 portfolio misses remain open.

The N4 wiring tests are green, but the scheduling-parity ledger explicitly keeps combined verification pending. In particular, the public GF(3) Gaussian, Direct PHP, and Tseitin actions have typed allowances, exact replay, raw effort/yield feedback, and disabled default rows, but no accepted interval, order, dose, or default admission. Several older ledger rows also retain cumulative acceptance obligations.

Next actions and risks

The correctness prerequisite is effective Default/random fuzz showing real producer calls, offers, applications, replay, and independent proof/model checking for every enabled structured family. The handoff and immutable sanitizer pin are prepared; no external owner campaign is counted until it is actually launched and returns a report.

When benchmark execution is authorized, first run only the repaired `cabp`, Break, and Timetable trajectory controls. Require the accepted structured prefix before interpreting wall time. If restored, evaluate the saved target projections and then B28+C5, Rank1–50, WV100, and WV2–89 under the 1,800-second checker cap and completed-proof slowdown rule. The full goal still requires all 44 independently checked misses and those acceptance gates.

The detailed evidence is the repository's current audit and complete Debug checkpoint.